**Predictability Effects in Natural Reading are Logarithmic, Not Linear: Evidence from an Eye-Movement Replication of Brothers and Kuperberg (2021)**

How long it takes to read a word depends on the word's predictability given its prior context (e.g. 1, 2). The *form* of this relationship—whether reading time is linear or logarithmic in relation to contextual predictability—is of substantial theoretical importance, as it arbitrates between theories of the predictability effect. A linear relationship is predicted if the effect is due to pre-activation of highly predictable words (e.g. 3, 4), while a logarithmic relationship is predicted by Surprisal theory (5,6). While several corpus-based studies have obtained evidence for a largely logarithmic relationship using various methodologies (e.g. 6, 7, 8), Brothers & Kuperberg (9; henceforth "BK21") found evidence for a linear relationship across a three-word region in a high-powered self-paced reading (SPR) experiment with well-controlled stimuli, where the same target word appeared with different levels of predictability across three English sentences. The goal of the present study was to assess the form of the relationship with the same stimuli, but using the more naturalistic eyetracking-while-reading paradigm in place of SPR. In our experiment, subjects (N=86) each read 216 sentences, with each sentence representing one item in a triplet wherein a critical, target word was embedded in a low-, moderate-, or high-predictability sentence context. An example appears below.

For our initial analysis, we created linear mixed-effect models (LMEs) to predict first fixation duration (FFD) and gaze duration (GD) on the target word. We also conducted an exploratory analysis of word skipping, though we note that neither theory makes a clear prediction about the form of the relationship between predictability and word skipping. These models varied in whether they used a word's raw, untransformed predictability estimate, or a log-transformed value to predict reading times. We also compared BK21's original cloze values vs. estimates generated by GPT-2 (12), as previous research has indicated that the source of these estimates may affect the form of the relationship (13). We used Akaike weights (10, 11) to compare the relative evidence in support of each model. Across all analyses, we found overwhelming support for logarithmic models compared to linear alternatives, and models based on GPT-2 predictability values outperformed their cloze counterparts; models using log-transformed GPT-2 predictability had by far the most support. Figures 1 and 2 show the condition-level and item-level relationship between FFD and GD respectively, and Table 1 provides an overview of our results.

Further analyses used log-transformed dependent measures, as is typical in eye-movement research (e.g., 14) and models that included both linear and logarithmic terms (c.f. 7). In each analysis, we found consistent support for logarithmic models using GPT-2 predictability estimates. We also investigated the existence of potential spill-over effects, confirming that the effect of a target word's predictability on eye movement measures is almost entirely localized to the word itself, rather than appearing on subsequent words, as BK21 had found in their SPR experiment.

Overall, we found strong support for Surprisal's prediction of a logarithmic relationship between predictability and reading time. This runs counter to BK21's original finding, but is consistent with most other research on the topic. We speculatively attribute the BK21 result to the use of cloze probability to estimate the predictability of their target words (see 13 for further discussion of this issue) and the nature of SPR as a paradigm, but our experiment nonetheless provides convincing evidence for a logarithmic relationship using the very same stimuli.

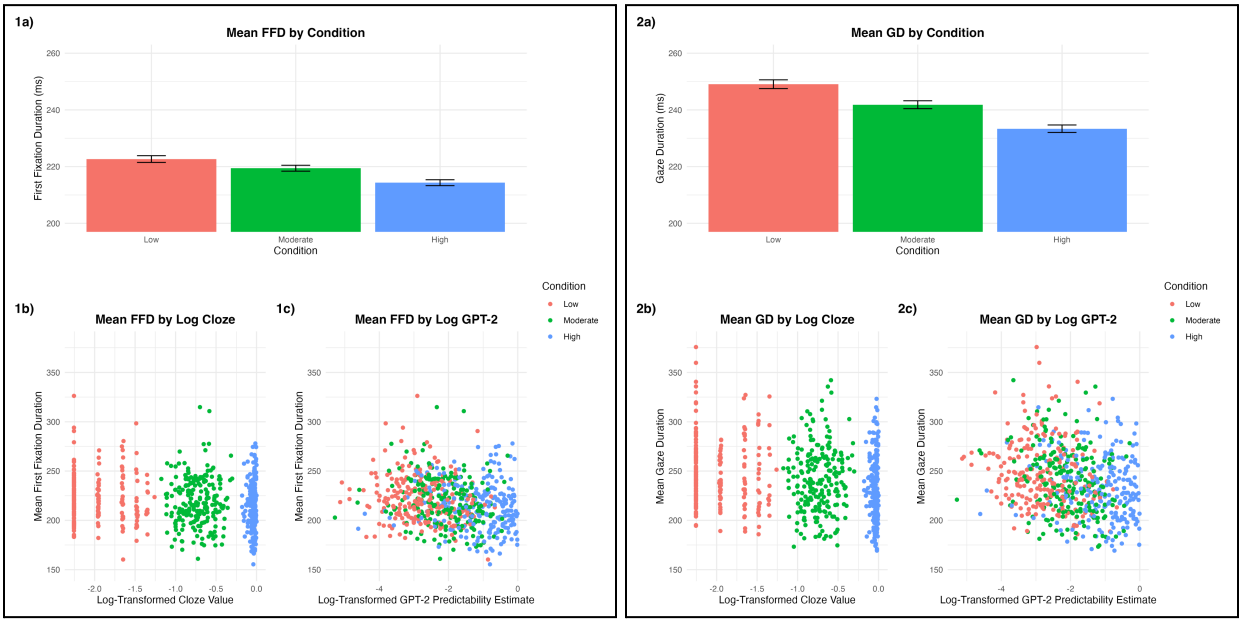
[1] West & Stanovich, 1978. *Child Development.* [2] Ehrlich & Rayner, 1981. *Journal of Verbal Learning & Verbal Behavior.* [3] Reichle, Rayner, & Pollatsek, 2003. *The Behavioral and Brain Sciences.* [4] Rayner, Reichle, Stroud, & Williams, 2006. *Psychonomic Bulletin & Review.* [5] Levy, 2008. *Cognition*. [6] Smith & Levy, 2013. *Cognition.* [7] Szewczyk & Federmeier, 2022. *JML.* [8] Goodkind & Bicknell, 2018. *Proceedings of the 8th Workshop on Cognitive Modeling and Computational Linguistics.* [9] Brothers & Kuperberg, 2021. *JML.* [10] Wagenmakers & Farrell, 2004. *Psychonomic Bulletin & Review*. [11] Akaike, 1973. *Proceedings of the Second International Symposium on Information Theory*. [12] Radford et al., 2019. *OpenAI Blog*. [13] Shain et al., 2024. *PNAS*. [14] Burchill & Jaeger, 2024. *JML*.

## Example Item

a. *They knew that Harry kept her awake by* ***barking*** *through the night.* (low pred.)
b. *Our pets sometimes ask us for treats by* ***barking*** *through the night.* (moderate pred.)
c. *The neighborhood dog always kept her awake by* ***barking*** *through the night.* (high pred.)

## Figures 1 and 2.

*Mean FFD by Condition (1a), Item-Level Log-Transformed Cloze Values (1b), and GPT-2 Probability Estimate (1c); Mean GD by Condition (2a), Item-Level Log-Transformed Cloze Values (2b), and GPT-2 Probability Estimate (2c)*




## Table 1.

| Model | Models Using Cloze Values | | Models Using GPT-2 Estimates | |
|---|---|---|---|---|
| Eye-Movement Metric | AIC of Linear Model (Akaike weight) | AIC of Logarithmic Model (Akaike weight) | AIC of Linear Model (Akaike weight) | AIC of Logarithmic Model (Akaike weight) |
| First Fixation Duration | 158,073.1 (0.000) | 158,069.2 (0.002) | 158,079.3 (0.000) | 158,056.6 (0.998) |
| Gaze Duration | 165,313.3 (0.001) | 165,307.9 (0.021) | 165,344.2 (0.000) | 165,300.2 (0.978) |
| Skipping | 15,401.4 (0.016) | 15,401.3 (0.017) | 15,406.7 (0.001) | 15,393.2 (0.966) |